Customer Shopping Behaviour Analysis

Uncovering insights from **3,900 transactions** across demographics, product preferences, and subscription behaviour to guide strategic business decisions.

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Dataset at a Glance

3,900

Transactions

Rows of purchase data

18

Features

Columns of attributes

654

Customers

Unique buyers analyzed

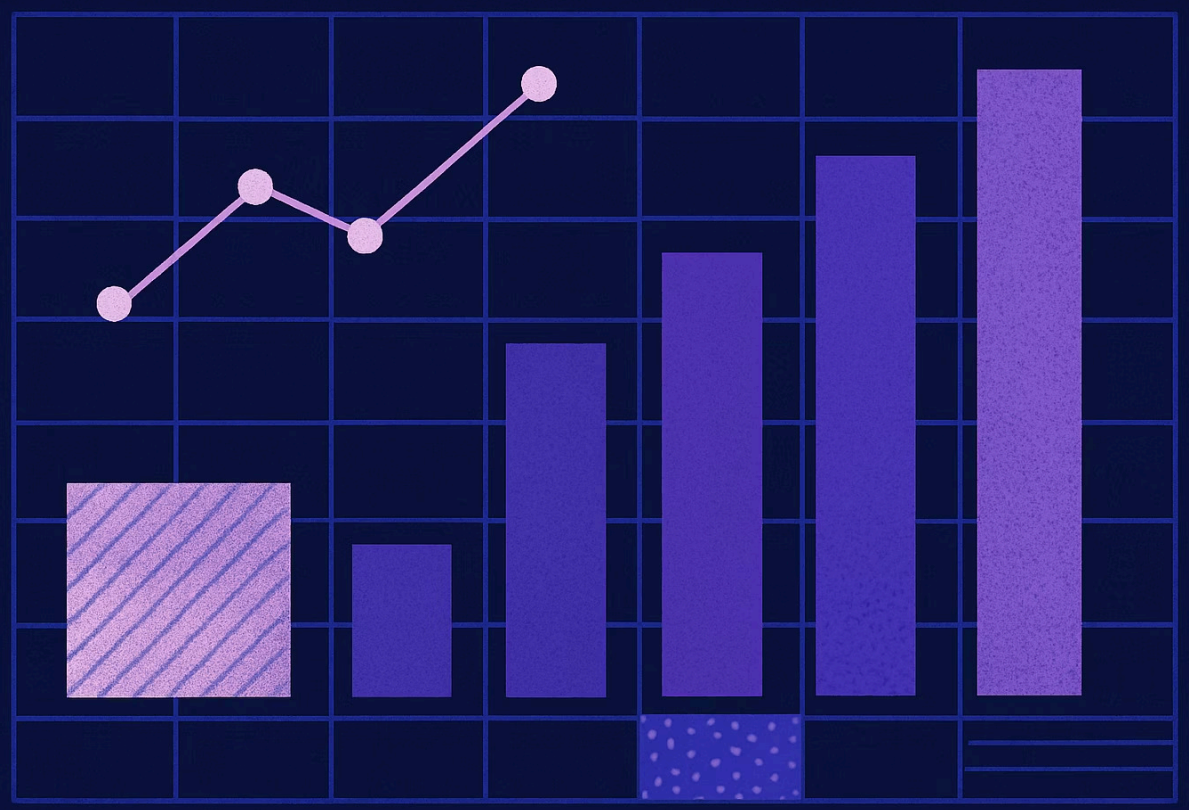
37

Missing Values

In Review Rating column

Key Features

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Colour
- **Behaviour:** Discount, Promo Code, Frequency, Review Rating, Shipping Type



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Data Preparation in Python

01

Data Loading & Exploration

Imported dataset via pandas; used `df.info()` and `df.describe()` for structure and summary stats.

02

Missing Data Handling

Imputed 37 missing Review Rating values using the **median rating per product category**.

03

Feature Engineering

Created `age_group` by binning ages; derived `purchase_frequency_days` from purchase data.

04

Database Integration

Connected Python to **PostgreSQL** and loaded the cleaned DataFrame for SQL analysis.

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SQL Analysis: Revenue & Customer Spend

Revenue by Gender

Compared total revenue generated by male vs. female customers.

High-Spending Discount Users

Identified customers who used discounts but still spent **above average** purchase amount.

Shipping Type Comparison

Compared average purchase amounts between **Standard** and **Express** shipping.

Revenue by Age Group

Calculated total revenue contribution of each age group.

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SQL Analysis: Products & Ratings

Top 5 Products by Rating

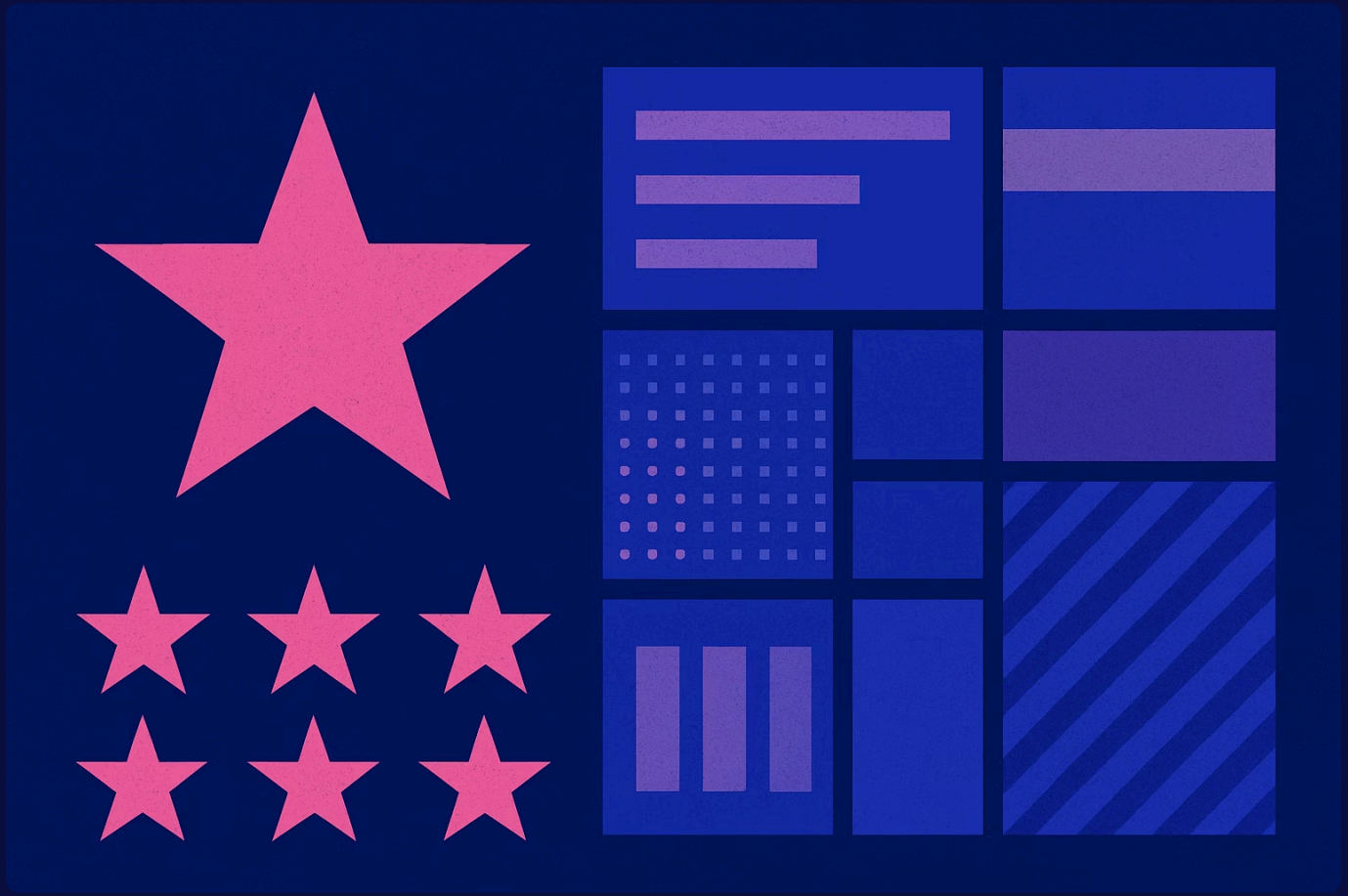
Found products with the highest average review ratings.

Discount-Dependent Products

Identified **5 products** with the highest percentage of discounted purchases.

Top 3 Products per Category

Listed the most purchased products within each category.



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Customer Segmentation & Subscriptions

1

New Customers

First-time buyers with limited purchase history.

2

Returning Customers

Repeat buyers showing growing engagement.

3

Loyal Customers

High-frequency buyers — prime subscription targets.

Key Finding: Customers with **>5 purchases** were analyzed for subscription likelihood — repeat buyers are significantly more likely to subscribe.

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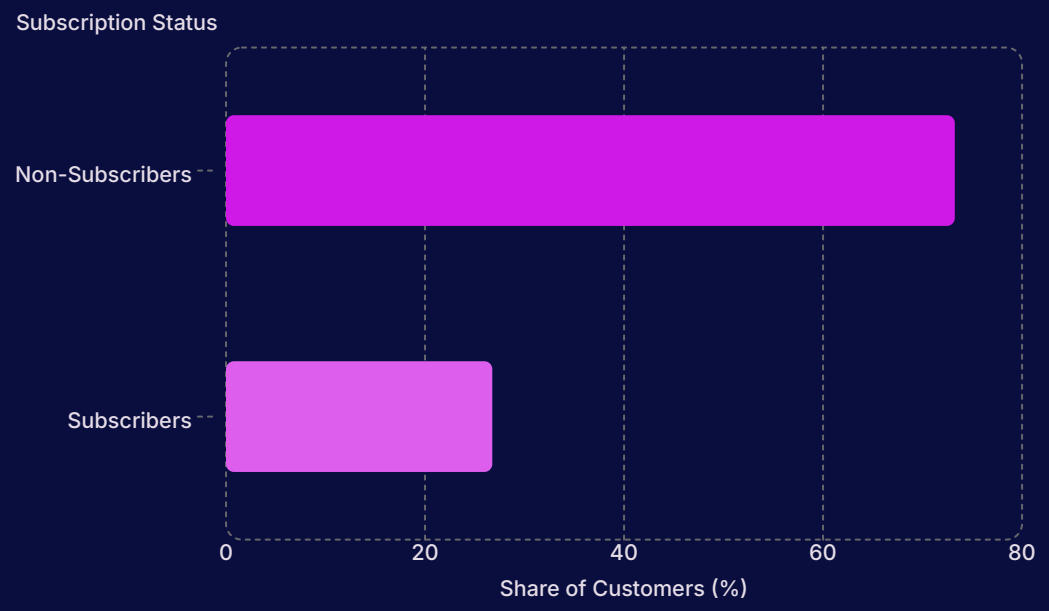
Subscribers vs. Non-Subscribers



The Subscription Gap

A striking **73.24%** of customers are non-subscribers, while only **26.76%** hold a subscription. Average spend and total revenue were compared across both groups to quantify the opportunity.

This represents a major growth opportunity — targeted campaigns could significantly increase subscription adoption.

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Power BI Dashboard: High-Level Metrics

$38.23K

Total Revenue

Across all transactions

$58.46

Avg. Purchase

Per transaction

3.82

Avg. Rating

Out of 5 stars

654

Customers

Analyzed in dashboard

Dashboard Features

- Filters by **shipping type** (2-Day, Express, Free) and **season** (Fall, Spring, Summer, Winter)
- Treemap of item-level purchases: **Skirts, Shorts, T-shirts, Sneakers, Jeans, Handbags**
- Gauge chart tracking satisfaction (range: 2.5–5.0)

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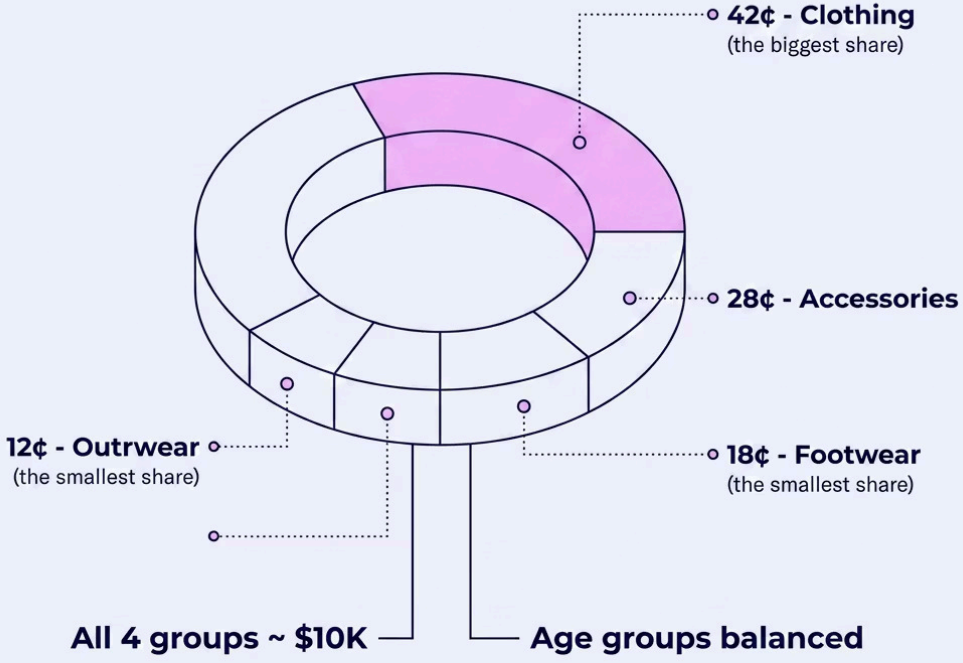
Key Product & Demographic Insights



What the Data Reveals

- **Clothing** dominates revenue vs. Accessories, Footwear, and Outerwear
- Revenue is **fairly balanced** across Youngsters, Adults, Middle-aged, and Senior citizens — each contributing ~$10K
- Broad demographic appeal — **no single age group dependency**
- Average rating of **3.82/5** indicates room for improvement in customer experience

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to close the 73% non-subscriber gap.

Customer Loyalty Programs

Reward repeat buyers to accelerate movement into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control — especially for discount-dependent products.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI.

Product Positioning

Highlight top-rated and best-selling products in campaigns to drive conversions.

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